

Exp no 4 : Strategize impedance matching with characteristic impedance and load impedance through the measurement of a) Location of the stub b) Length of the stub

Test case 1:

```
clear;

clear;

clc;

ZR=100;Zo=600;f=100*(10^6);

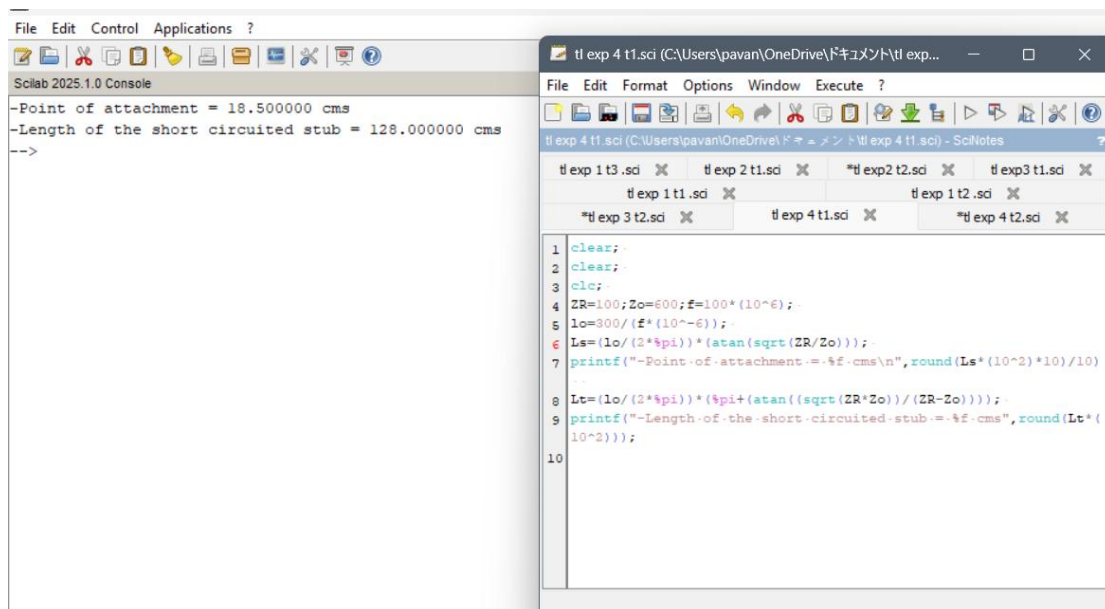
lo=300/(f*(10^-6));

Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));

printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)

Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));

printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```



Test case 2 :

```
clear;
```

```
clear;
```

```
clc;
```

```
ZR=100;
```

```
Zo=600;
```

```
f=100*(10^6);
```

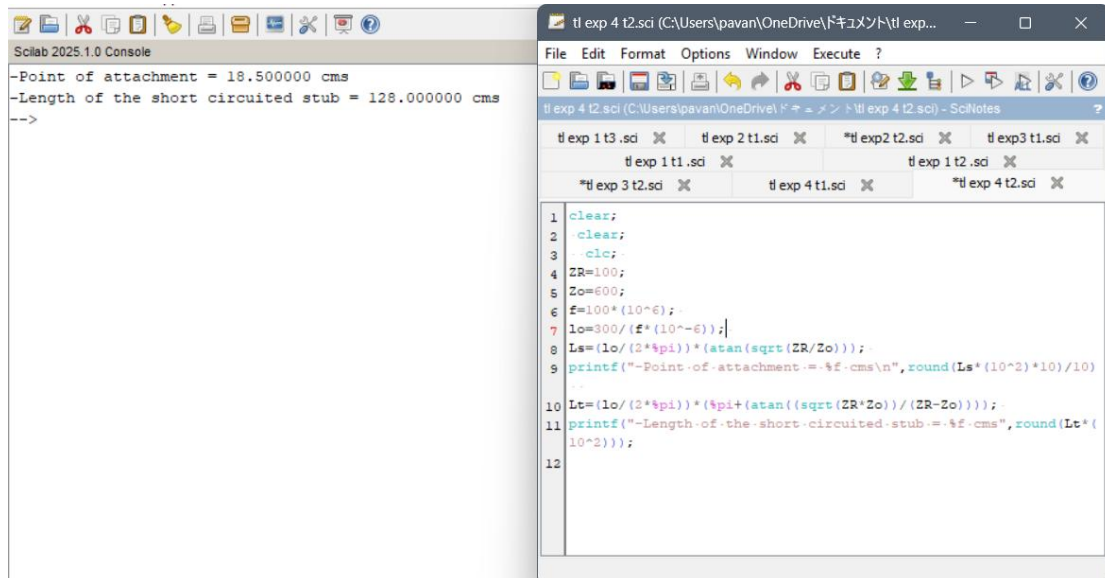
```
lo=300/(f*(10^-6));
```

```
Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
```

```
printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
```

```
Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
```

```
printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
```



The screenshot displays the Scilab 2025.1.0 environment. The left pane shows the console output, and the right pane shows the script editor with the code being executed.

Scilab 2025.1.0 Console

```
-Point of attachment = 18.500000 cms
-Length of the short circuited stub = 128.000000 cms
-->
```

tl exp 4 t2.sci (C:\Users\pavan\OneDrive\ドキュメント\tl exp...)

```
File Edit Format Options Window Execute ?
tl exp 4 t2.sci (C:\Users\pavan\OneDrive\ドキュメント\tl exp 4 t2.sci) - SciNotes
tl exp 1 t3.sci X tl exp 2 t1.sci X *tl exp 2 t2.sci X tl exp 3 t1.sci X
tl exp 1 t1.sci X tl exp 1 t2.sci X
*tl exp 3 t2.sci X tl exp 4 t1.sci X *tl exp 4 t2.sci X

1 clear;
2 clear;
3 clc;
4 ZR=100;
5 Zo=600;
6 f=100*(10^6);
7 lo=300/(f*(10^-6));
8 Ls=(lo/(2*%pi))*(atan(sqrt(ZR/Zo)));
9 printf("-Point of attachment = %f cms\n",round(Ls*(10^2)*10)/10)
10
11 Lt=(lo/(2*%pi))*(%pi+(atan((sqrt(ZR*Zo))/(ZR-Zo))));
12 printf("-Length of the short circuited stub = %f cms",round(Lt*(10^2)));
13
```